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REVIEW

Bacterial disease in seaweed aquaculture: pathogen diversity and future solutions

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Abstract

In recent years, there has been a rapid expansion in seaweed aquaculture due to the increased demand for sustainable food, pharmaceuticals, and other industrial needs. This growth has been accompanied by a rise in disease prevalence, jeopardising the economic viability of seaweed farming and the livelihoods of millions reliant on the sector. Here, through focusing on the complex interactions between seaweed hosts and their microbial communities, we review current knowledge on bacterial seaweed pathogens. We discuss the challenges associated with pathogen identification, including the constraints of traditional approaches and the complexities arising from resident commensal bacteria. We then assess the use of probiotics, selective breeding, Integrated Multi-Trophic Aquaculture (IMTA), site selection, and the priming of early life stages as emerging disease management strategies, while highlighting the need for omics-based studies to identify disease biomarkers and commercially relevant disease-resistant strains. We conclude by underlining the importance of early detection systems and improved knowledge dissemination within the sector to enable a more proactive and sustainable approach to disease management.

Keywords Infectious algal diseases · Bacterial pathogens · Macroalgae · Beneficial bacteria · Future solutions · Aquaculture · Marine

Introduction

Seaweeds (macroalgae) are a diverse group of photosynthetic organisms that inhabit coastal regions worldwide (Lüning 1991). Although cultivated historically, seaweed aquaculture has intensified since the 1950s and has been actively pursued for use in the food, nutraceutical, pharmaceutical, and cosmeceutical industries (FAO 2018; Lomartire et al. 2021). Recently, the term "phyconomy" has been proposed to encompass this large-scale, economically beneficial rise in seaweed aquaculture (Hurtado et al. 2019). The sector currently accounts for 52% of mariculture (Chopin and Tacon 2021; FAO 2024), is valued at US$14–15 billion (World Bank 2023), and makes use of over 240 commercially cultivated species, only ten of which are extensively farmed (Hermans 2021; FAO 2024).

Aside from industrial uses, cultivated seaweeds also provide ecological benefits, such as serving as nurseries and breeding grounds for crustaceans and fish (Hasselström et al. 2018), reducing eutrophication (Fei 2004), and aiding in the bioremediation of heavy metals (Davis et al. 2003) and organic contaminants (Hardegen et al. 2023). Seaweed aquaculture is also hypothesised to offer other benefits such as biofuel production, climate mitigation through assimilating CO_2 and carbonate dissolved in seawater during growth, as biofertilisers for soil enrichment providing organic matter and nutrients, and decreasing cattle methane emissions when incorporated into livestock feeds and supplements (Duarte et al. 2017; Roque et al. 2021). However, these uses all require full life-cycle and mass balance accounting to effectively establish any genuine greenhouse-gas reductions due to emissions generated in the production process (Hurd et al. 2022; Troell et al. 2022; Pessarrodona et al. 2024).

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Since the beginning of the century, expansions in commercial seaweed production have led to elevated levels of diseases and pests, with major production nations encountering substantial disease outbreaks in both wild and cultivated populations (FAO 2018). In some cases, these outbreaks have resulted in the loss of up to 50% of seaweed biomass (Ward et al. 2020; Spagnuolo and Genovese 2024; Zhang et al. 2024) and 30–50% of total farming costs (Kim et al. 2014). As millions from coastal communities rely on income from this sector (FAO 2020), disease outbreaks have significant social and financial repercussions. For instance, in the Philippines alone, outbreaks of ice-ice disease (IID), which affects tropical seaweeds such as *Kappaphycus* and *Eucheuma,* caused an estimated annual loss of US$100 million (Ward et al. 2020, 2022). Climate change has also been shown to increase the range, prevalence, and severity of bacterial diseases, while simultaneously compromising host metabolism and immune responses (Harvell 1999; 2002; 2009; Tracy et al. 2019). To develop future-proof disease management strategies, it is therefore crucial to have a comprehensive understanding of the diversity of current bacterial diseases, the factors determining disease susceptibility of hosts and possible preventative measures.

Previous reviews have focused on the varied effects of invertebrate, oocyte, virus, protist, and epiphyte diseases in seaweeds (Largo 2002; Egan et al. 2014; Egan and Gardiner 2016; Ward et al. 2020, 2022; Behera et al. 2022; Spagnuolo and Genovese 2024). Here we assess diseases caused by bacterial pathogens as they form an emerging threat for both cultivated and wild populations and pose unique management challenges due to their genetic plasticity and complex relationship with the host. We begin by giving a broad overview of the functioning of healthy seaweeds and the known diversity of seaweed bacterial pathogens. We place particular emphasis on the role commensal bacteria play in disease progression, highlighting how environmental changes can cause resident bacterial species to shift from beneficial to pathogenic roles, shaping the onset and severity of disease. We next review current and emerging solutions in seaweed disease management, focusing on the protective role of the microbiome and up-and-coming technologies.

The seaweed holobiont

Through producing organic compounds which act as nutrients and chemo-attractants, seaweeds create metabolic niches (de Oliveira et al. 2012), which are utilised by a variety of microorganisms, some of which form symbiotic relationships with the host, establishing as an integrated, functional unit, otherwise known as the holobiont (Egan et al. 2013; Simon et al. 2019; Saha et al. 2024; Qui-Minet et al. 2025). The primary colonisers of this holobiont are usually bacteria, owing to their rapid growth rates and capacity to metabolise algal secretions (Wahl 1989; Lachnit et al. 2009). Once established, bacterial symbionts play essential roles in nutrient acquisition (Croft et al. 2005; Wahl et al. 2012), the induction of growth and morphogenetic factors (Provasoli and Pintner 1980; Matsuo et al. 2003, 2005), and in deterring fouling organisms and pathogen colonisation via the production of antifouling (Armstrong et al. 2001; Dobretsov and Qian 2002) and antimicrobial compounds (Zheng et al. 2005). The production of these secondary metabolites can also increase the host's resilience to environmental stressors by mitigating oxidative damage and host immune responses (Dittami et al. 2016; Ghaderiardakani et al. 2020) and thus further contribute to disease resistance (Li et al. 2022a). Consequently, seaweed health is often linked to the host's ability to recruit and regulate bacteria necessary for successful functioning under adverse conditions (Kessler et al. 2018; Ghaderiardakani et al. 2020; Hmani et al. 2024).

How similar microbiomes are across different individuals, specifically, whether individuals share a 'core microbiome' remains under debate (Vos 2023). Core microbiomes have been identified in *Gracilaria vermiculophylla* across seasonal scales (Mudlaff et al. 2025) and in *Laminaria hyperborea*, *Saccharina latissima* and *Durvillaea* spp. (King et al. 2023; Pearman et al. 2024) across spatial scales, however the proportion of core taxa appears to be low. This may, at least in part, reflect functional redundancy, whereby different taxa perform similar metabolic roles (Vos 2023).

To exert control over their microbiome and combat disease, seaweeds possess a range of innate immune defences (Fig. 1) (Weinberger 2007). For example, seaweeds such as *Laurencia dendroidea* and *Delisea pulchra* have been shown to express a broad spectrum of gene transcripts associated with the biosynthesis of defensive compounds such as terpenoids and halogenated furanones, indicating a constitutive, non-specific response to their microbiome members (Manefield et al. 2011; de Oliveira et al. 2012; Lang et al. 2024). Seaweeds therefore seem to follow the "ecosystem on a leash" paradigm in which their microbiome is strongly influenced by host control (Wilde et al. 2024). Through the production of antimicrobial compounds, seaweeds can regulate their microbiome by impeding bacterial settlement (Maximilien et al. 1998), growth (Maximilien et al. 1998; Dworjanyn et al. 2006), quorum sensing (González and Keshavan 2006; Harder et al. 2012) and weakening pathogen virulence (Da Gama et al. 2014; Egan et al. 2014). Consequently, seaweeds can be highly selective toward bacteria, a selectivity that can be further influenced by the antimicrobial action of the bacteria themselves (Wiese et al. 2009).

Seaweeds possess both constitutive barriers and inducible innate responses to infection (Weinberger 2007). Unlike the cell-based immune systems seen in other organisms, seaweeds use microbe-associated molecular patterns (MAMPs) and pathogen-induced molecular patterns

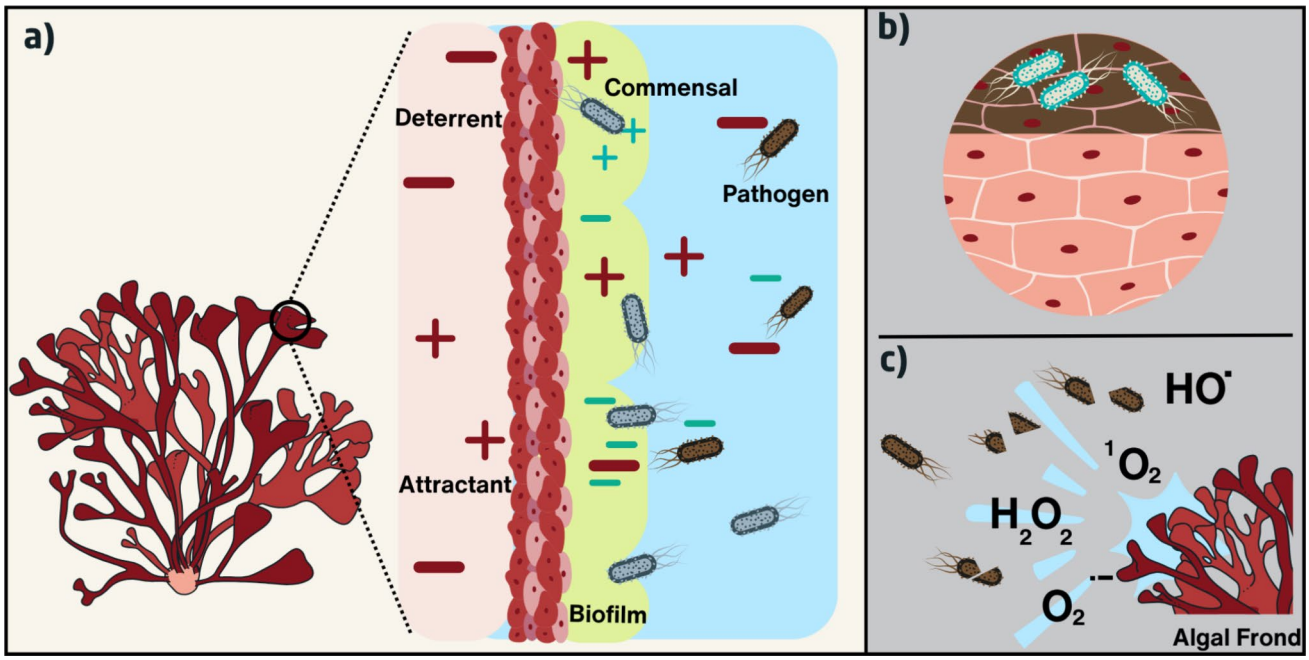


Fig. 1 Innate Defence Mechanisms of the Seaweed Holobiont. Seaweeds have evolved a variety of innate constitutive and inducible defences against bacterial infection. **a** Constitutive holobiont defences include the production of chemical compounds by the seaweed host and/or its associated microbial symbionts, which deter potential pathogens or attract other microbiome members. **b** The hypersensitive response, an inducible form of programmed cell death, limits pathogen spread adjacent to the infection site. **c** Oxidative bursts, another inducible innate response involving the rapid release of reactive oxygen species (ROS) that can inhibit pathogens or activate downstream defence pathways

(PIMPs) to differentiate beneficial and pathogenic microbes (Zipfel and Felix 2005; Weinberger 2007). Upon detection of these molecular signals, seaweeds initiate signal transduction cascades that increase the expression of defence-related proteins, energy conversion, and the biosynthesis of reactive oxygen species (ROS) (Küpper et al. 2001, 2009; Weinberger 2007; de Oliveira et al. 2017). These pathways ultimately activate key innate defence mechanisms, such as oxidative bursts, cell shedding, and hypersensitive responses at infection sites (Weinberger 2007; Li et al. 2024).

Reactive oxygen species (ROS), released during oxidative bursts, are frequently used by seaweeds to counter bacterial infections through exerting cytotoxic effects on invading pathogens or by triggering further defensive responses (Weinberger 2007). Similarly, by using hypersensitive responses, seaweeds can limit the spread of pathogens by destroying cells adjacent to infection sites (Weinberger and Friedlander 2008). Although this method is effective, if used excessively, the host's susceptibility to disease can increase due to either pathogen resistance or physiological damage to the host (Egan et al. 2014). This is often exacerbated in environments such as high-density aquaculture, where seaweeds experience increased pathogen exposures and repeated immune activations. These conditions increase the cumulative cost of host defence, compromising host health while increasing the risk of disease.

The diversity of bacterial pathogens of seaweeds

Although many studies have reported disease symptoms or syndromes in seaweeds (Craigie and Correa 1996; Gachon et al. 2010), relatively few have conclusively identified specific causative pathogens, mechanisms of action, or possible environmental risk factors. Establishing causality typically relies on frameworks such as Koch's postulates, which require the pathogen to be present in all instances of disease, isolatable in pure culture, able to induce identical disease symptoms in healthy specimens and be re-isolated from these infected hosts (Koch 1891). In addition, misattribution of disease symptoms is common, especially when disease expression varies markedly between hosts (Largo 2002). For instance, a single pathogen might induce rapid tissue necrosis in one host species under certain environmental conditions, whilst causing only mild discolouration or no visible symptoms in other species or environments (Largo et al. 1995a, 1995b; Gachon et al. 2010). Due to the diverse and largely unculturable nature of microbial communities, possible asymptomatic or polymicrobial nature

of infection, and experimental challenges, these postulates are usually not applied to seaweeds.

Distinguishing specialised pathogens from more opportunistic species remains a major challenge in the study of seaweed diseases (Egan et al. 2013). While specialised ('primary') pathogens can initiate disease in healthy hosts, opportunistic pathogens are usually commensal, causing disease only when the hosts are compromised. Opportunistic pathogens can further be distinguished from secondary colonisers which do not initiate disease but rather colonise decaying tissue (Egan et al. 2013). A well-documented example involves bacteria such as *Flavobacterium*, *Pseudoalteromonas*, and *Vibrio*, which were once thought to cause green-spot disease in *Pyropia* spp. (Sunairi et al. 1995) but are in fact secondary colonisers following a viral infection by PyroV1 (Kim et al. 2016). To account for the difficulty in applying a consistent pathogen definition, we here follow the pragmatic approach taken by Bartlett et al. (2022) designating seaweed pathogens when they are isolated from a symptomatic infection and considered to be the causative agent by the reporting researchers.

The majority of bacterial pathogens implicated in seaweed disease belong to the *Alteromonas, Pseudomonas, Pseudoalteromonas* and *Vibrio* genera within the Pseudomonadota (formerly gamma-Proteobacteria) phylum (Table 1). Many of these taxa are already recognised as pathogens in other aquaculture systems, such as the Pacific white-leg shrimp (Alfiansah et al. 2020; Arunkumar et al. 2020). It is clear that commercial interest has been a main driver of pathogen identification. Consequently, most studies focus on members of the Rhodophyta inhabiting tropical regions, with extensive investigations into genera such as *Eucheuma* and *Kappaphycus* due to the widespread prevalence and economic impact of ice-ice disease (Table 1; Largo 2002). In contrast, few bacterial pathogens have been identified in temperate climes. Of note is the apparent absence of confirmed bacterial pathogens affecting Chlorophyta hosts, which likely reflects the underrepresentation of this phylum among cultivated seaweeds (Nayar and Bott 2014).

Knowledge of disease outbreaks in natural populations lags behind that in aquaculture, even though the consequences can be substantial both ecologically and economically. For example, a decrease in primary production and carbonate accretion occurred due to an outbreak of coralline lethal orange disease (CLOD) affecting a vast population of reef-building coralline algae in the early 1990 s (Littler and Littler 1994, 1995). Consequently, wild populations not only have to contend with naturally occurring disease outbreaks but repercussions of the rapidly expanding aquaculture sector, such as the introduction of non-native diseases, epidemics, and a decrease in the genetic diversity of natural stocks (Spagnuolo and Genovese 2024).

Notably, fewer than half of the reported seaweed diseases here are associated with a single bacterium, highlighting the limitations of the classic "one pathogen = one disease" model (Vayssier-Taussat et al. 2014). Historically, studies have primarily concentrated on monomicrobial infections, but more recently many infections have been shown to be polymicrobial either in origin or manifestation (Tay et al. 2016). Pathogens often live in complex microbial communities where different interactions determine the progression of disease (Chow et al. 2011; Rogers 2012). This is the basis of the "pathobiome" concept, in which pathogenicity is conditional, modulated by interactions with the host, environmental factors and fellow microbiome members (Bass et al. 2019).

Commensal bacteria: Friend or Foe?

Although infectious disease is often associated with invading pathogens, emerging evidence indicates that many diseases appear, not from external pathogens, but resident and usually benign members of the host's microbiota (Chow et al. 2011; Egan and Gardiner 2016). For instance, *Staphylococcus aureus* is usually a commensal component of the human microbiome yet can cause severe and sometimes fatal infections when host defences are compromised (Turnbaugh et al. 2009). Seaweeds are no exception, with ten of the 23 pathogens listed in Table 1 previously associated with beneficial traits in other seaweed species.

Environmental stress events often coincide with disease outbreaks in both wild and cultivated seaweeds (Campbell et al. 2011; Ward et al. 2022). For example, *Vibrio alginolyticus* strain X-2 can upregulate host genes linked to stress and immune responses and preserve epibacterial community structure in *S. japonica,* while causing disease in a diverse array of other seaweed hosts (Ma et al. 2023). Through using -omics techniques, disease states have been shown to be associated with shifts in microbial gene expression and abundance rather than the appearance of entirely new taxa (Fernandes et al. 2012; Egan et al. 2013). This suggests that pathogenicity is highly context-driven, dependent on the expression of virulence traits under certain conditions. In the plant literature, the term "eubiosis" refers to a normally functioning, homeostatic microbiome associated with a healthy individual (Paasch and He 2021), where as a disruption is often referred to as "dysbiosis" and is widely known to detrimentally impact host health (Li et al. 2022a; Ling et al. 2022). However, it must be noted that this term remains contentious due to the absence of a clear definition and its assumption of an 'ideal' microbial state (Hooks and O'Malley 2017; Mancuso et al. 2023).

In line with the phenotypic plasticity of many marine bacteria, the ability to shift from commensal residents to pathogens is unsurprising. Numerous taxa possess latent

Table 1 An overview of the diversity of suspected bacterial pathogens of seaweeds, including their recorded hosts, confounding variables and possible mechanisms of action. Suspected pathogens were only included if bacteria were identified to at least genus level and were able to establish the same disease symptoms upon reintroduction to healthy specimens; the concentration of inoculum is also stated. This table excludes descriptive studies or those investigating community-level changes within diseased specimens without interrogation of Koch's postulates. Host species belonging to wild populations are denoted with an asterisk (*)

Disease	Identified Pathogen	Identified Seaweed Hosts	Location	Symptoms	Risk Factors	Experimental Inoculation Conc	Comments	References
Suminori Disease	*Actinomycetota: Arthrobacter tumbae DY1219-4Y* *Bacteroidota: Gaetbulibacter saemankumensis H-14LY, H-15LY & H-16LY*	*Pyropia yezoensis*	Japan	Black lustreless colouration and plasmoptysis of thalli cells, reduces economic value	> Low temperatures > Low salinity	$1.6 - 8.3 \times 10^7$ CFU mL^{-1}	Mitigation: Exposure of nets to air and acid washing at pH2. Bacteriophage "U2" can repress *Gaetbulibacter* sp. H-14LY	Kusuda et al. (1992); Mine et al. (2009); Mine et al. (2010)
Anaaki Disease (Alternative Names: *Pin-Hole, Green Spot & Green Rot Disease)*	*Bacteroidota: Flavobacterium* sp. *LAD-1*	*Pyropia yezoensis*	Japan, Korea	Pin holes appear at the centre of the thallus, gradually widening into lesions with wide green borders, causing fragmentation of the thallus. Slimy rots, holes in blades, cession of growth may also appear. Degradation of the whole plant can occur as quickly as 1–2 days	> Low temperatures > Surface wounds > Ammonia availability	1×10^7 CFU mL^{-1}	Mitigation: Acid dipping infected thalli and exposure of nets to air and acid washing Mechanism: Bacteria with hydrolytic activity toward porphyrin, β-D-galactosidase, carboxymethyl cellulase, xylanase, protease and agarase activity	Fujita (1990); Sunairi et al. (1995); Kim et al. (2014)
	Pseudomonadota: Pseudomonas sp. *JY38*	*Porphyra dentata*	Korea			1×10^8 cells mL^{-1}		Park et al. (2001)
	Unspecified bacterium "*DOR*" isolate	*Chondrus crispus*	Canada			Cultures were "quite turbid"		Craigie and Correa (1996)
Rotten Thallus Syndrome (Alternative Names: *Thallus Whitening & Bleaching Stipe Disease)*	*Pseudomonadota: Pseudoalteromonas gracilis B9*	*Gracilaria gracilis*	Philippines, South Africa	White to pink discolouration and gradual disintegration of thallus. Erosion of the pericarp, revealing the cortical and medullary cells	> Reduced water flow > High temperatures	1×10^9 CFU mL-1	Mitigation: Bacteria sensitive to polymyxin B, nalidixic acid, nitrofurazone and oxytetracycline Mechanism: Plasmid pDA1 responsible for agarolytic activity	Lavilla-Pitogo (1992); Schroeder et al. (2003)
Tip Bleaching/ Whitening Stripe Disease	*Pseudomonadota: Vibrio parahaemolyticus, Thalassospira* spp.	*Gracilariopsis lemaneiformis*	China	Thallus whitening and apical necrosis, disintegration of thallus	> High culture density > Low light intensity > Low salinity	40,000 cells mL^{-1}	Mitigation: 1-octen-3-one, E-2-nonenal, and benzaldehyde thought to be characteristic metabolites of infection and could be used as biomarkers for the disease	Sun et al. (2012)

Table 1 (continued)

Disease	Identified Pathogen	Identified Seaweed Hosts	Location	Symptoms	Risk Factors	Experimental Inoculation Conc	Comments	References
White Tip Disease	Unspecified Bacterium Strain *OR-11* *Bacteroidota: Flavobacterium-Cytophaga* complex *Actinomycetota: Corynebacterium-Arthrobacter* complex	*Gracilaria conferta*	Israel	White necrotic tips followed by disintegration of thallus	> High temperature > High light intensity > High culture density > Poor aeration	2.11×10^7 cells mL^{-1}; 1:8554 ratio of cells	Mitigation: Disease could be prevented with Rifampicin, Vancomycin and Cefotaxim. Other epibacteria could also prevent bleaching	Weinberger et al. (1994), Weinberger (1997); (1999)
Bleaching Disease	*Pseudomonadota: Nautella* sp. *R11, Phaeobacter* sp. *LSS9, Alteromonas* sp. *BL110, Agarivorans* sp. *BL7* *Bacteroidota: Aquimarina* sp. *AD1 & BL5*	*Delisea pulchra**	Australia	Line of bleaching from infection site. Loss of pigment causes lower photosynthetic, reproductive abilities and susceptibility to grazing, tissue necrosis and death	> High UV > High temperatures	1×10^6–10^7 CFU mL^{-1} 1×10^6 cells mL^{-1}	Mechanism: *Nautella* sp. *R11* penetrates epidermis and cortical cells causing necrosis. Pathogenicity is helped by possessing adhesion mechanisms, algal metabolite transport systems, enzymes to resist oxidative stress, cytolysins, LuxR gene varR and RaiR regulator for coordinating virulence factors. Other bacteria have strong agarase activity, phycobiliprotein can also decrease on infection	Case et al. (2011); Fernandes et al. (2011); Gardiner et al. (2015a); (2015b); Kumar et al. (2016); Hudson et al. (2018)
	Pseudomonadota: Agarivorans albus *Bacteroidota: Aquimarina latercula* *Actinomycetota: Brachybacterium* sp.	*Gracilaria lemaneiformis*	China			1×10^7 CFU mL^{-1}		Liu et al. (2019)
Scraping White Disease ((Alternative Names: *Bleaching Disease & White Strip Disease)*	*Pseudomonadota: Phaeobacter italicus* JN-W1	*Porphyra* sensu lato *conchocelis*	China	White spots, strips or patches on shells and lesions and white patched on thalli		1×10^7 CFU mL^{-1}	Mitigation: Pathogen can be inhibited under low temperature and pH Mechanism: Infection is contact dependent	Liu et al. (2025)

Table 1 (continued)

Disease	Identified Pathogen	Identified Seaweed Hosts	Location	Symptoms	Risk Factors	Experimental Inoculation Conc	Comments	References
Coralline Lethal Orange Disease (CLOD)	*Bacillota: Planococcus citreus, Planococcus* sp., *Planococcus mcmeekinii, Bacillus pumilus* *Pseudomonadota: Pseudomons oleovorans*	Crustose algae, *Porolithon* spp.*	Cook islands, Fiji South Pacific reefs, Hawaii)	Occurs as a front of bright orange dots that move across the host, leaving behind a bleached carbonate skeleton	> High temperatures	Not provided	Mitigation: High wave energy habitats may inhibit attachment of CLOD propagules	Littler and Littler (1994), (1995); Cervino et al. (2005); Aeby (2007)
Yellow Spot Disease	*Pseudomonadota: Vibrio mediterranei* 117-T6	*Pyropia haitanensis conchocelis*	China	Yellow spots appear on infected shell-borne conchocelis sporeling stage which spread into lesions	> High temperatures > High light intensity > Low pH	6×10^7 CFU mL^{-1}	Mechanism: Bacterium secretes algicidal substances	Yang et al. (2020); Xu et al. (2020)
Macular Disease/ Pseudomonas-like Bacteriosis	*Pseudomonadota: Pseudomonas*-like bacteria	*Porphyra* spp.	Japan	Small red/brown spots on conchoecelis of sporelings before turning yellow/white. Turns *Porphyra* gametophytes green	> High temperatures	1×10^7 CFU mL^{-1}	Mitigation: Discarding infected shells, chlorine treatments or increasing water flow, some sensitivity to nifurstyrenate and erythromycin	Yamanoi and Takami (2008)
Gall Formation	*Pseudomonadota: Roseobacter* group	*Prionitis lanceolata; Prionitis filiformis; Prionitis decipiens**	North & South America	Gall like tumours consisting of hypertrophied algal cells		Not provided		Ashen and Goff (1996, 2000)

Table 1 (continued)

Disease	Identified Pathogen	Identified Seaweed Hosts	Location	Symptoms	Risk Factors	Experimental Inoculation Conc	Comments	References
Ice-Ice Disease (IID)	*Pseudomonadota: Vibrio* sp. *P11, Vibrio* sp. *ABI-TU15, Vibrio alginolyticus, Alteromonas macleodii, Aurantimonas coralicida, Shewanella haliotis, Stenotrophomonas Maltophilia, Brucella anthropic, Catenococcus thiocycli, Pseudomonas aeruginosa* *Bacteroidota: Cytophaga* sp. *P25-Flavobacterium* complex *Actinomycetota: Arthrobacter nicrotiannae* *Bacillota: Bacillus subtilis*	*Kappaphycus alvarezii*	Philippines, Indonesia, India, Japan, Malaysia	Whitening and hardening or softening of tissue leading to disintegration. There can also be mucus discharge	> High light intensity > High temperatures > Low salinity > Stressed hosts > Low water movement > High phosphate concentrations > Epiphyte infestations	1×10^3–10^8 cells mL^{-1} 1×10^6 CFU mL^{-1}	Mitigation: Farm seaweeds in areas with strong water exchange, *Vibrio* ABI-TU15 growth can be inhibited by *Halomonas* species Mechanism: Hydrolytic enzymes utilised to penetrate medulla, causing epidermal degradation and destruction of pigmented plasmids	Uyenco et al. (1980); Largo et al. (1995a, (1995b), (1999); Achmad et al. (2016); Irmawati and Sudjiro (2017); Syafitri et al. (2017); Azizi et al. (2018); Zainuddin et al. (2019); Abdul Malil (2022)
	Pseudomonadota: Vibrio sp. *P11* *Bacteroidota: Cytophaga* sp. *P25-Flavobacterium* complex	*Eucheuma denticulatum*	Philippines			1×10^3–10^5 cells mL^{-1}		
	Pseudomonadota: Pseudomonas spp., *Vibrio* spp., *Chromobacterium* spp., *Achromobacter* spp. *Bacteroidota: Flavobacterium* spp.	*Gracilaria verrucosa*	Indonesia			1×10^6 CFU mL^{-1}		
	Pseudomonadota: Vibrio owensii	*Halymenia floresii**	Mexico			Addition of 30 µL of cultured bacteria		
Bleaching Disease	*Pseudomonadota: Pseudoalteromonas piscicida X-8, Alteromonas* sp. *A-1, Alteromonas espejiana*	*Saccharina japonica*	China	Bleached white tips, swollen gametophyte cells, filamentous bleaching, loss of thalli. Blades bleach and deteriorate	> High temperatures	1–3 × 10^8 CFU mL^{-1}	Bacteria can produce alginate lyase, invade the endodermis causing cell corruption and death	Wang et al. (2006); Peng and Li (2013); Zhang et al. (2022)

Table 1 (continued)

Disease	Identified Pathogen	Identified Seaweed Hosts	Location	Symptoms	Risk Factors	Experimental Inoculation Conc	Comments	References
Green rot disease/ Falling Off disease	*Pseudomonadota:* Alginic acid-decomposing bacteria *Pseudomonas* sp.	*Saccharina japonica*	China	Green colouration and softening of stipe, decay leads to detachment of plant	> Insufficient light	Not provided	Mitigation: Addition of erythromycin, discard decayed blades, maintain cultures at 6–8 ºC and replacing seawater	Wu (1990); Liu et al. (2002); Wang et al. (2014)
Thallus Bleaching disease	*Pseudomonadota: Alteromonas* sp.	*Saccharina religiosa*	Japan	Induced lesions and thallus bleaching	> Low salinity > High temperature	3×10^8 cells mL^{-1}		Vairappan et al. (2001)
Red spot disease	*Pseudomonadota: Pseudoalteromonas bacteriolytica* sp.	*Saccharina japonica*	Japan	Red spotted staining of culture beds and detachment of sporophyte	> High temperatures > High POM concentrations > High levels of bacteria flora	Not provided	Mitigation: Washing "Mother" seaweed with sterilized seawater and cutting off decayed parts	Ezura et al. (1988); Sawabe et al. (1998); Yumoto et al. (1989a), (1989b)
Hole-Rotten Disease	*Pseudomonadota: Pseudoalteromonas* sp., *Vibrio* sp., *Halomonas* sp.	*Saccharina japonica*	China	Green rotting on sporophytic edges and holes on blade, which spread until detachment from ropes		1×10^7 CFU mL^{-1}		Wang et al. (2008)

virulence factors which, although tightly repressed under normal conditions, can be upregulated under host stress or environmental fluctuations (Fernandes et al. 2011; Billaud et al. 2022; Bharathan et al. 2025). For seaweeds, which lack an adaptive immune system, even subtle changes in environmental conditions can disrupt the host's surface chemistry or microbial-mediated checks which suppress virulence, permitting formerly harmless commensals to proliferate and initiate disease (Zozaya-Valdes et al. 2015).

One of the best-documented examples of dysbiosis in seaweeds is the bleaching disease of the red alga *Delisea pulchra.* In this system, changes in microbial community composition can occur prior to visible tissue damage (Fernandes et al. 2012). Several bacterial taxa usually present in the microbiome, including *Nautella italica* R11 and *Phaeobacter galleciensis* LSS9, can cause disease when host defence mechanisms are compromised, such as at elevated temperatures, often due to increases in their abundance (Case et al. 2011; Fernandes et al. 2011; Gardiner et al. 2017). Following this, the bacteria deploy virulence factors and QS-dependent regulatory mechanisms, exploiting the weakened host (Case et al. 2011; Fernandes et al. 2011; Gardiner et al. 2017). Similarly, when exposed to low light intensity and salinity, resident bacteria of *Kappaphycus alvarezii,* such as *Vibrio* sp. P11 and *Cytophaga* sp. P25, can induce ice-ice disease (Largo et al. 1995b).

The difficulty in identifying seaweed pathogens thus stems from the fact that many are not obligate pathogens but rather commensal microbiota 'gone rogue', shifting their behaviour to exploit weakened hosts. In the context of seaweed aquaculture, where monocultures can exacerbate microbial imbalance by permitting the rapid spread of pathogens, predicting and mitigating disease requires identifying the conditions under which commensal bacteria shift from benign association to causing opportunistic disease. Although challenging, this diagnostic complexity highlights the need for adopting a pathobiome perspective or framework that recognises that disease is a result of many complex interactions between the host, microbiome, and the wider environment (Foster et al. 2017; Bass et al. 2019). Future disease control, therefore, is most likely to make advances where it moves towards manipulating the microbial environment rather than targeting individual pathogens.

Future directions in seaweed disease control

Due to the many health benefits of the seaweed microbiome, recent efforts to enhance host growth, environmental resilience, and disease resistance through scalable and cost-effective microbial approaches have increased in both agricultural and conservation contexts (El-Saadony et al. 2021; Peixoto et al. 2022; Voolstra et al. 2024). In the next part of this

review, we explore key developments in aquaculture practises and emerging disease management research topics for this expanding sector (Fig. 2). While nanotechnology, antibiotics and acid-washing seaweed blades have shown some potential such as reducing fungal disease in rainbow trout eggs (Johari et al. 2016) or manipulating epiphytic communities to enhance seaweed biomass (Blanco et al. 2025), we do not address their application here, as the open nature of seaweed farming poses too great a risk of environmental contamination, especially when more suitable alternatives are available.

Direct microbiome manipulation

Probiotics The use of bacteria with antagonistic activity against pathogens has frequently been used to enhance the resilience of microbiomes associated with terrestrial crops (Quiza et al. 2015). For example, reducing the severity of rice blast disease by up to 70% can be achieved through enriching native bacteria such as *Streptomyces* sp. UPMRS4 and *Pseudomonas mosselii* BS011 (Awla et al. 2017; Wu et al. 2018). Such techniques could be beneficial in seaweed cultivation, where several bacteria have already been identified to increase disease resistance by enhancing host health (Li et al. 2022b; Ma et al. 2023). Although, in contrast to terrestrial crops, the application of these bacteria (also known as Seaweed Beneficial Microorganisms, SBMs (Li et al. 2023)) in seaweed aquaculture remains relatively novel.

Beneficial bacteria may prevent disease through several mechanisms: either by directly inhibiting pathogen virulence (Fan et al. 2020), monopolising resources (Gu et al. 2020), interacting with other microbiome members (Li et al. 2022a), stimulating host bioactive compound production (Wen et al. 2023), or producing antimicrobials (Granato et al. 2019). For instance, two strains of *Pseudoalteromonas piscicida,* P3 and P6, were shown to inhibit disease symptoms inflicted by *Pythium porphyrae* in *Pyropia yezoensis,* providing an effective biocontrol methodology against Red Rot Disease (Weng et al. 2024). Similarly, *Phaeobacter* sp. BS52 protects *Delisea pulchra* against the opportunistic bleaching pathogen *Aquimarina* sp. AD1 (Li et al. 2022b). Interestingly, this species was also found to diminish the detrimental effects of bleaching disease when inoculated onto non-host species (Li et al. 2022b). As bleaching diseases are likely to become more common, this probiotic bacterium might be promising in enhancing host disease resistance in future ocean warming scenarios.

Two methods for probiotic application are currently being explored: either the inoculation of non-native SBMs or the reinoculation of an isolated and enriched native bacterium (Lau et al. 2022). A third possibility is the introduction of genetically modified strains, but this would require strict regulation (Ke et al. 2021). Regardless of method, further work is needed to ascertain whether introduced bacteria could be maintained in large-scale industrial settings and, if so, how they transfer between host life stages.

Environmental and biosecurity risks also require attention. Threats such as the unintended emergence of novel diseases need to be prevented, either through limiting the use of microbes to secure hatchery stages or through solely applying the microbe's beneficial active compound, for instance, directly applying the compound thallusin rather than its bacterial producers (Dhiman et al. 2022; Ulrich et al. 2022). Similarly, prebiotics, non-digestible feed additives historically used to improve gut microbiome health in fish, can enhance nutritional uptake and resistance to disease (Dimitroglou et al. 2011; Ganguly et al. 2013). In seaweeds, such additives could be used in a similar manner, either to nourish beneficial bacteria or as potential pathogen deterrents. The enrichment of beneficial host microbiomes has already been observed under land-based cultivation and laboratory conditions thought to be mediated by intrinsic host factors such as Algal Growth and Morphogenesis-Promoting Factors (Ghaderiardakani et al. 2019; Califano et al. 2020; Nguyen et al. 2023). Seaweeds rely on metabolites with antifoulant and antimicrobial activity: for example, compounds such as halogenated furanones and fucoxanthin can mitigate surface fouling and bacterial attachment (Campbell et al. 2011; Saha et al. 2011), and 1-octen-3-one, E-2-nonenal, benzaldehyde, and oxylipins can counter bacterial disease (Sun et al. 2012). However, research into seaweed prebiotics is still in its infancy and it is still unclear whether beneficial compounds can be effectively applied or upregulated within host tissues.

Selective breeding Selective breeding, which utilises controlled reproduction of individuals possessing desirable characteristics, has already been implemented for various economically important seaweeds, such as *Porphyra* and *Laminaria* species (Chaoyuan and Guangheng 1987; Dai et al. 1993). To produce characteristics such as enhanced taste and resistance to high irradiance, temperatures and tissue rot disease, hybridisations such as those of *Saccharina longissima* and *S. japonica* have also been utilised (Li et al. 2007; Robinson et al. 2013; Hwang et al. 2019).

Although widely used in terrestrial agriculture, omics techniques such as metabolomics and genomics could also be applied to seaweed cultivation (Patwary et al. 2021). These techniques could be used to mitigate disease by identifying genes, regulatory elements and metabolic pathways linked to resistant traits, including those integral to microbiome regulation, antimicrobial compound production and adaptive immune responses (DeWeese and Osborne 2021). Such genes can be incorporated into breeding programmes

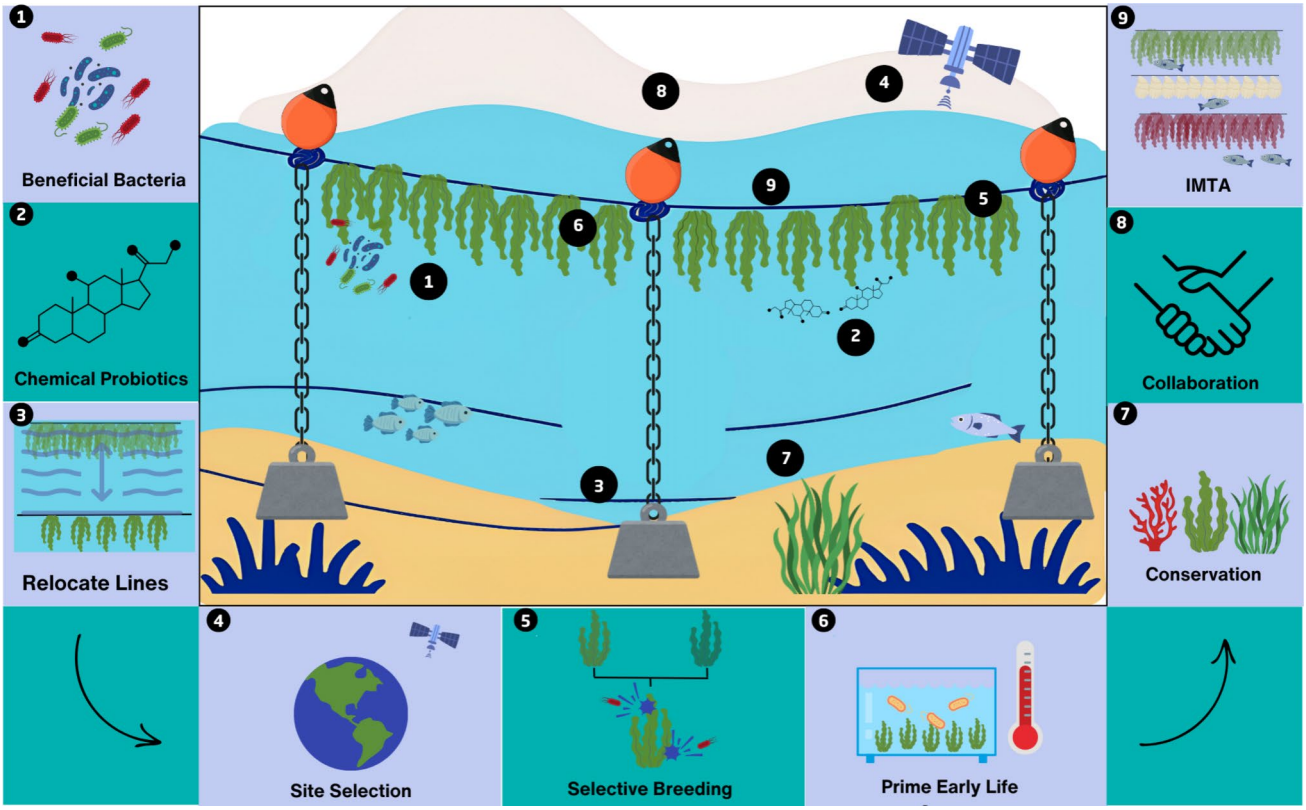


Fig. 2 Future solutions for disease mitigation in aquaculture include (1) manipulating the microbiome to become more resilient with beneficial bacteria. (2) Using probiotics to increase host health. (3) Changes to infrastructure such as lowering cultivation lines to lower water depths in intense UV or temperatures. (4) Modelling future disease outbreaks and favourable climate conditions to optimise site selection. (5) Selectively breeding disease-resistant strains. (6) Priming early life stages to be more resilient. (7) Conserving the genetic diversity of wild populations. (8) Increased collaboration between scientists, farmers and industry to disseminate pertinent knowledge more efficiently. (9) The use of IMTA methods to reduce disease susceptibility

through marker-assisted selection or introduced via genetic engineering.

As of 2024, over 110 seaweed genomes representing 105 species are publicly available, aiding new selective breeding programmes (Wang et al. 2020; Nelson et al. 2024). These datasets enable the application of Genome-Wide Association Studies (GWASs), which explore single-nucleotide polymorphisms (SNPs) associated with desirable qualities, and Quantitative Trait Loci (QTL) mapping, which can detect the genetic loci contributing to specific phenotypes (Murray et al. 2008; Huang and Han 2014; Huang et al. 2015; DeWeese and Osborne 2021; Feng et al. 2023). Through combining these approaches, candidate genes and genomic regions linked to disease-resistant traits could be identified and incorporated into genomic predictions to develop disease-resistant seaweed strains.

To identify future commercial cultivars, transcriptomics could be used to highlight gene expression profiles connected to beneficial traits (Chin et al. 2017; Thien et al. 2020; Zhuang et al. 2024). By comparing strain-specific transcriptomes exposed to stress, beneficial traits such as elevated growth rate, disease resistance and higher carrageenan content could be identified and used to support selection and breeding decisions. For example, a substantial reservoir of differentially upregulated defensive genes, such as heat shock proteins, secondary metabolites, protective enzymes, and cellulase and protease inhibitors has already been found in *P. yezoensis* (Tang et al. 2019). Likewise, transcriptomic activation of genes connected with stress response pathways and cell-wall strengthening has been observed in a disease-resistant genotype of *S. japonica* (Zhuang et al. 2024). This approach could be further extended to investigate traits which aid the maintenance of beneficial interactions within the host's microbiome.

While classical breeding is an effective strategy, its success is inherently restricted to traits already present in the breeding population. Genetic engineering, already used in several seaweed species, permits the introduction of traits not naturally present, such as the expression of recombinant proteins and reporter genes to produce antigens, antibodies and other therapeutical compounds (Jiang et al. 2002; Zhang et al. 2008; Badis et al. 2021; Shen et al. 2023; Trujillo

et al. 2024). Farms growing genetically engineered disease-resistant strains may be operational soon, depending on their adherence to GMO and biosecurity policies. However, even though research is rapidly progressing in this area, many nations lack clear regulations governing genetically modified or selectively bred seaweed strains (Goecke et al. 2020).

Priming early life stages Priming early life stages to sublethal stresses activates specific stress-responsive genes and can be used to increase resilience within cultivars (Jueterbock et al. 2021). For example, in *A. esculenta* sporophytes, exposure in early cultivation stages to high light intensity reduces thermal stress by enhancing photoprotective mechanisms (Martins et al. 2022). In other brown macroalgae, similar priming effects have been noted where thermal or oxidative stress causes persistent changes in antioxidant capacity or defence-related gene expression (Heinrich et al. 2012).Therefore, similar to abiotic stressors, exposing seaweeds to microbial and chemical cues could prime the host against subsequent pathogenic challenges.

Instead of applying direct stressors, waterborne chemical cues released by certain herbivores may stimulate the production of defensive compounds and bolster seaweeds against disease outbreaks (Toth and Pavia 2000). As seaweeds are known to utilise a broad spectrum of chemical cues to attract beneficial bacteria and suppress disadvantageous strains (Saha and Weinberger 2019), such metabolites could also be explored for potential use in recruiting beneficial bacteria. While early-life priming may offer a novel approach to mitigating disease in future commercial cultivars, there still is a lack of evidence that this strategy increases disease resistance in macroalgae. Further experimental work is needed to assess if this approach could indeed increase the defensive capacity of seaweeds against pathogenic infection.

Practices that promote microbial eubiosis

Site selection and farming practises To reduce disease outbreaks, a combination of technological advancements and improved aquaculture practises, such as suitable site selection, are needed. For example, during periods of heightened light intensity and temperature, farms could relocate lines to deeper waters, decreasing the chance of thermally induced diseases (Makame et al. 2021). Similarly, improving water flow by reducing overcrowding could aid the dispersal of beneficial microbes while simultaneously inhibiting the accumulation of opportunistic pathogens, thereby decreasing the transmission of pathogens and promoting resilient microbial communities.

However, for sustainable, long-term cultivation, effective site selection must also incorporate broader spatial areas than individual farms. To assess locations for optimal environmental parameters for both the seaweed host and its microbiome, increasingly sophisticated climate and oceanographic modelling is now available (Veenhof et al. 2024). Bathymetric and hydrodynamic analysis of a previously failed site in Lampung Bay, Indonesia, exemplifies the need for rigorous analysis in the placement of future ventures (Yulianto et al. 2017). Similar investigation of areas of low water flow or proximity to freshwater sources may also increase the likelihood of success, reducing stress and microbial disruption caused by salinity fluctuations and increased pathogen loads (Largo 2002; van der Loos et al. 2022).

Socio-economic and practical considerations are as valuable as environmental factors when assessing cultivation sites. Studies from Zanzibar and Europe have revealed the importance of aligning farming sites with the farmer's local preferences, access to infrastructure and the area's traditional uses (Hedberg et al. 2018; Thomas et al. 2019). Overlooking these factors may result in the abandonment or underutilisation of sites, regardless of environmental suitability. Looking ahead, the co-location of cultivation sites with offshore wind parks could be a possible solution for sharing marine space, reducing user conflicts and benefiting from the sheltering effect of existing infrastructure (Coffey et al. 2025). The integration of socio-economic data with oceanographic modelling is therefore essential for future spatial planning of farming ventures in changing climates.

Co-culturing and Integrated Multi-Trophic Aquaculture (IMTA) In coastal settings, the expansion of invertebrate and fish aquaculture can cause significant shifts to water quality due to the release of elevated nutrients (Gao et al. 2021), disrupting ecosystem functioning and contributing to the spread of disease. Approaches which co-culture several species, such as *Kappaphycus* spp. with *Eucheuma denticulatum,* have already been shown to enhance resilience and decrease the prevalence of diseases such as IID in summer months (Pang et al. 2015). A meta-analysis revealed that proximity of seaweed farms to seagrass fields reduces disease risk (Fiorenza et al. 2024). It is possible that co-cultivation of commercial seaweed crops with specific other 'companion crop' seaweed species could equally reduce disease incidence (e.g. Fu et al. 2015), but to our knowledge this has not been tested.

The creation of a balanced ecosystem through cultivating different trophic levels, known as integrated multi-trophic aquaculture (IMTA), represents a logical step in future disease management, as multiple species provide enhanced disease resistance and help to curtail environmentally harmful practises (Troell et al. 2009; Kim et al. 2014). Seaweeds

absorb nitrates, ammonia, phosphates and heavy metals, acting as natural bio-remediators; their inclusion in other aquaculture systems can therefore mitigate nutrient pollution while maintaining economic value (Gao et al. 2021; Sugumaran et al. 2022). Beyond disease mitigation, these seaweeds can provide habitat and improve water quality, ultimately benefiting higher trophic levels by enhancing food availability and providing new breeding and nursery habitats (Corrigan et al. 2024).

Knowledge sharing Disease outbreaks in intensive aquaculture systems are usually multifactorial and may result from the disregard of region-specific cultivation guidelines, limited knowledge on symptoms and pathologies, low genetic variation causing decreased resilience of populations, poorly chosen cultivation sites, and/or the introduction of novel pathogens through farming non-native species (Spagnuolo and Genovese 2024). Addressing these issues requires improved collaboration among researchers, industry stakeholders, and policymakers to promote the dissemination of knowledge and best practices. Such coordination could be addressed at scientific meetings (e.g., the International Society for Applied Phycology; the International Seaweed Symposium organised by the International Seaweed Association and Seagriculture EU and Asia–Pacific), but digital tools and citizen science programmes could also play key roles. For example, the wider adoption and maintenance of platforms used to support the early detection and management of diseases, such as the web portal "My Seaweed Looks Weird", which enabled farmers to report unusual diseases and submit samples (Strittmatter et al. 2022), could improve disease resistance and resilience across the sector.

As reservoirs of genetic diversity, wild seaweed populations offer resilience to the emerging threats of climate change and therefore must be restored and conserved (Eger et al. 2024). The mapping of genetic variation among populations with valuable cultivation and restoration traits is already underway (Mao et al. 2020). Combined with the establishment of biobanks, these approaches may preserve genetic material and create a safeguarded reserve for future biodiversity and breeding efforts (Cottier-Cook et al. 2016; Wade et al. 2020; Brakel et al. 2021; Fouqueau et al. 2024; Hofmann et al. 2025), supporting the development of robust, climate-resilient aquaculture systems.

Finally, implementing a systematic health assessment framework, such as that proposed by Work and Aeby (2006) for coral disease, could standardise disease diagnosis between regions. Through integrating microbiological and environmental contexts and providing consistent terminology and symptom documentation, such frameworks could reduce losses in aquaculture through improving health monitoring throughout the entire seaweed life cycle while enhancing differentiation between healthy and diseased specimens.

Conclusions

Seaweed aquaculture is rapidly expanding due to its role as a sustainable food source (both as whole seaweeds and extracts used in processed foods), its support of coastal livelihoods, and its contribution to the UN Sustainable Development Goals. However, disease outbreaks increasingly endanger the resilience of this sector. Our review highlights that maintaining seaweed health requires advancing beyond a single-pathogen view and embracing the holobiont perspective, acknowledging that seaweeds and their microbial communities function as a unified system.

A core concept in disease prevention is the dual role of commensal bacteria. Although microbes provide many beneficial services such as nutrient acquisition (Croft et al. 2005), defence (Armstrong et al. 2001), and stress tolerance (Dittami et al. 2016), under environmental stressors they can shift to pathogenic behaviour, complicating diagnosis. Therefore, maintaining microbial communities instead of controlling invading pathogens may provide a next logical step in disease management. Many emerging techniques support these views, with probiotics, selective breeding, and priming of early life stages able to enhance host resilience, while better site selection could reduce the impact of environmental stressors that cause dysbiosis. Early detection and subsequent targeted interventions may now be possible due to the increase in -omics techniques available to identify biomarkers of disease and resistant strains.

Nevertheless, serious challenges remain. The increasing pressures of climate change add to the urgency in developing dependable solutions and knowledge-sharing systems across the sector. To create long-term approaches for proactive disease prevention, seaweed aquaculture will need to depend on an integrated understanding of the holobiont including host biology, microbiome dynamics, and environmental change.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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